

Ventricular Tachycardia due to ARVD



Section I: Scenario Demographics

Scenario Title:	Ventricular Tachycardia due to Arrhythmogenic Right Ventricular Dysplasia
Scenario Developer(s):	Martin Kuuskne
Date of Development:	13/03/2015
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups

Section II: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To employ various ACLS algorithms in the setting of a rare, inherited myocardial disease that predisposes to ventricular arrhythmias.
CRM Objectives:	1) To ensure closed loop communication during ACLS treatment algorithms. 2) To demonstrate the ability to set dynamic priorities as a patient manifests deterioration.
Medical Objectives:	1) To employ the use of suitable antiarrhythmic medications or synchronized cardioversion in the setting of a stable, wide-complex tachycardia (WCT). 2) To recognize and rapidly respond to clinical deterioration in a patient requiring synchronized cardioversion and/or defibrillation.

Case Summary: A brief summary of case progression and major events
A 26-year-old man who suffered a syncopal event while playing soccer presents to the emergency department with a stable wide-complex tachycardia (WCT). The patient must be treated with an antiarrhythmic medication or by synchronized cardioversion. The patient later deteriorates into an unstable WCT and then ventricular fibrillation requiring advanced cardiac life support (ACLS) and defibrillation.

References
Burns, E. (2010, December 8). Sudden Syncope on the Soccer Field. Retrieved March 13, 2015, from http://lifeinthefastlane.com/ecg-exigency-008
Neumar R, et al. (2010) American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science. Circulation.2010; 122: S729-S767



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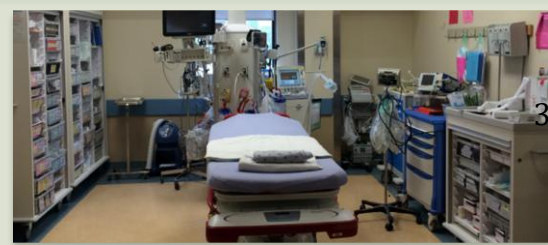


Section III: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case					
You are working an evening shift in a community hospital emergency department. A 26-year-old man presents to the ED by ambulance after an episode of syncope while playing soccer.					
B. Scenario Cast & Realism					
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual		
	☒ Mannequin		☐ Physical		
	☐ Standardized Patient		☐ Emotional/Experiential		
	☐ Hybrid		☐ Other:		
	☐ Task Trainer		☐ N/A		
		<i>Select most important dimension(s)</i>			
C. Required Monitors					
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line			
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography			
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:			
D. Required Equipment					
☒ Gloves	☒ Nasal Prongs	☐ Scalpel			
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit			
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit			
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit			
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit			
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit			
☐ Blood Products	☒ ET Tubes	☐ Other:			
☐ Intraosseous Set-up	☐ LMA	☐ Other:			
E. Moulage					
None required.					
F. Approximate Timing					
Set-Up:	5 min	Scenario:	12 min	Debriefing:	10 min



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Section IV: Patient Data and Baseline State

A. Patient Profile					
Patient Name: Peter Alvares		Age: 26		Weight: 70kg	
Gender: ☒ M ☐ F		Code Status: Full Code			
Chief Complaint: Syncope					
History of Presenting Illness: "I was playing soccer today and started to feel palpitations 10 minutes into the game; then I felt dizzy and my friends told me passed out for 10 seconds. Marco caught me before I hit the ground."					
Past Medical History:		None		Medications: None	
Allergies: NKDA					
Social History: EtOH 5-6 drinks on weekends. Denies illicit drugs, denies smoking.					
Family History: No family history of sudden cardiac death.					
Review of Systems:		CNS:		Normal	
		HEENT:		Normal	
		CVS:		Normal, no chest pain	
		RESP:		Normal, no SOB	
		GI:		Normal	
		GU:		Normal	
		MSK:		Normal	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 162/min		BP: 109/71		RR: 18/min O ₂ SAT: 99 %	
T: 36.8°C		CBGM: 6.5 mmol/L		GCS: 15 (E4V5M6)	
General Status: Alert, oriented, responding appropriately, speaking.					
CNS:		PEARL 2+, Normal EOM, no lateralizing signs in face/UE/LE. Normal reflexes.			
HEENT:		Normal			
CVS:		Tachycardia, no extra heart sounds.			
RESP:		GAEB, no adventitious breath sounds			
ABDO:		Soft, non-tender			
GU:		Normal			
MSK:		Normal		SKIN: Normal	

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Section V: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: V-Tach HR: 162/min BP: 109/71 RR: 18/min O ₂ SAT: 99% T: 36.8°C	A+Ox3 Pulse present No signs of shock, ischemic chest discomfort or acute heart failure.	Learner Actions ☐ Monitors/Full vitals ☐ Hx/PE ☐ IV access ☐ Supplemental O ₂ ☐ EKG #1 ☐ Recognizes <u>stable</u> wide complex tachycardia ☐ Antiarrhythmic (Procainamide/amiodarone) ☐ Pacer Pads placed ☐ Sedation (if cardioversion) ☐ Recap/Summary	Modifiers <i>Changes to patient condition based on learner action</i> Triggers <i>For progression to next state</i> - Antiarrhythmic OR Synchronized cardioversion → 2. Stabilization - AV nodal blockade (βB, CCB, Adenosine) or defibrillation → 3.Unstable
2. Stabilization Rhythm → Sinus HR → 75/min BP → 118/77	A+OX3 Pulse present	Learner Actions ☐ Labs ☐ CXR ☐ EKG #2 ☐ IV Fluids ☐ Antiarrhythmic infusion	Modifiers Triggers - 2 minutes → 3. Unstable
3. Unstable Rhythm → V-Tach HR → 180/min BP → 80/56	Altered mental status, Pulse present	Learner Actions ☐ Synchronized cardioversion (≥100J) ☐ Antiarrhythmic agent ☐ Defibrillation	Modifiers Triggers - Synchronized cardioversion → 4. V-Fib - Defibrillation → 4. V-Fib
4. V-Fib Rhythm → V-Fib HR → 200 BP → 0/0	Unresponsive	Learner Actions ☐ Chest compressions ☐ Immediate Defibrillation ☐ Epinephrine	Triggers 2 Defibrillations (2 cycles of CPR) → 5. Resolution
5. Resolution Rhythm → Sinus HR → 75 BP → 118/77 O ₂ SAT → 99%	A+Ox3, Pushing providers off of chest.	Learner Actions ☐ Repeat EKG (#1) ☐ Assesses if patient “follows commands” ☐ Repeat Vitals signs ☐ Recap/Summary ☐ Cardiology consultation	END SCENARIO PRN



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Section VI: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results
No Laboratory Results Given for this simulation.
Digital Images (EKGs, CXRs Etc.)
For CXR and EKGs, please refer to the case’s blog post at emsimcases.com .



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Section VII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How can you actively ensure closing the loop during ACLS? 2) What are suitable antiarrhythmics in the setting of a stable wide complex tachycardia as per ACLS? 3) What constitutes “unstable” in a patient with a wide complex tachycardia and a pulse? 4) What are the EKG findings of arrhythmogenic right ventricular dysplasia? 5) What is the difference between right ventricular outflow tract ventricular tachycardia and ventricular tachycardia associated with myocardial infarctions.	
Key Moments	
Recognition of stable ventricular tachycardia	
Recognition of clinical deterioration (“unstable”)	
Timely defibrillation	

